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A Appendix

Additional Figures

In this main paper, we report results based on models that are trained to optimize the $N = 1$ single token-ahead prediction, and we test those models for predictive accuracy for other N .

The same methods can also be used to optimize subsequent tokens, and the results of those methods are shown here. We find that optimizing for $N = 1$ works best and generalizes surprisingly well to other N , but that that optimizing for other N does not perform well for $N = 1$.

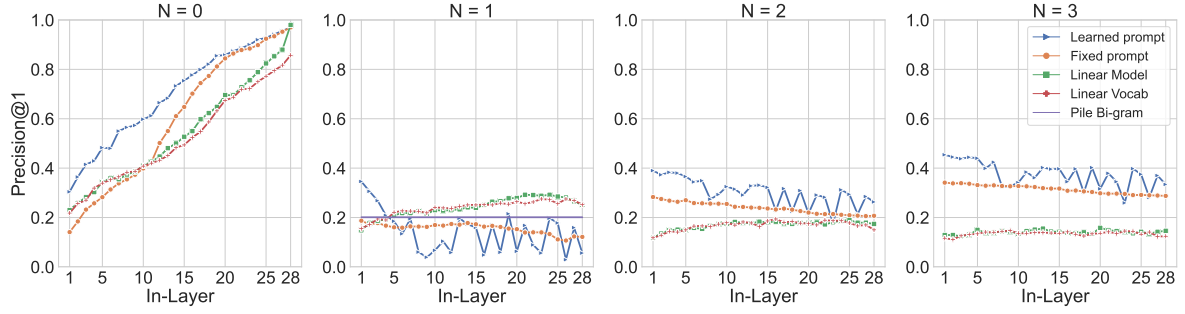


Figure 7: The Precision@1 (Accuracy) of all the methods trained with predicting the currently decoded token (teacher-forcing)

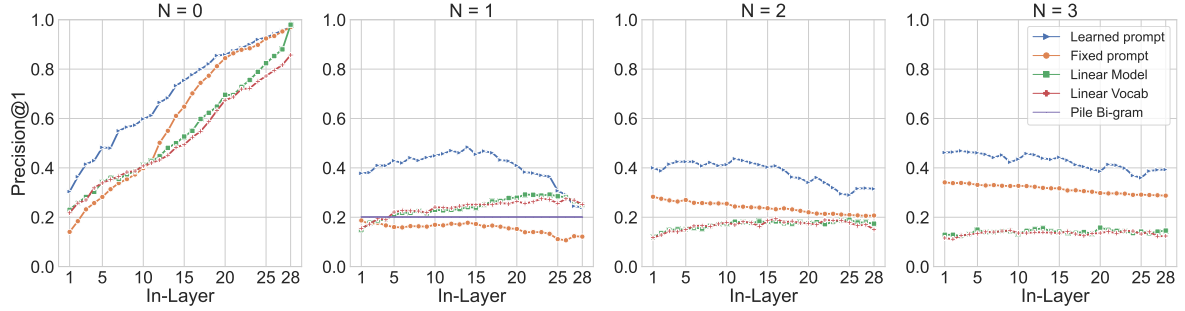


Figure 8: The Precision@1 (Accuracy) of all the methods trained with predicting the 1st next token

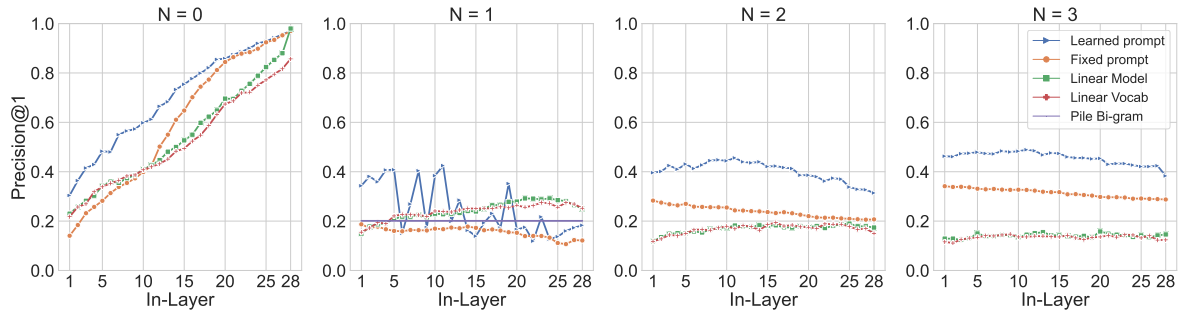


Figure 9: The Precision@1 (Accuracy) of all the methods trained with predicting the 2nd next token